gcloud CLI Commands

Category	Command	Tier	Added	Explanation	Example
Auth	`gcloud auth list`	1		Lists all credentialed accounts on the machine and marks the active one with an asterisk.	`gcloud auth list`
Config	`gcloud config list`	1		Displays the active configuration properties, such as project, account and region, or a single property if you name it.	`gcloud config list project`
Config	`gcloud config set project`	1		Sets the default project that subsequent gcloud commands run against.	`gcloud config set project agremotesensing`
Storage	`gcloud storage cp`	1		Copies files between local disk and buckets, supporting quoted wildcards for batch transfers.	`gcloud storage cp "gs://ag-remote-bucket01/MountainView_NDVI_2022*.tif" ~/Desktop/`
Storage	`gcloud storage ls`	1		Lists buckets or the objects under a bucket path, for example to confirm that exports landed.	`gcloud storage ls gs://ag-remote-bucket01/exports/`
Auth	`gcloud auth activate-service-account`	2		Authorizes gcloud to act as a service account by using its JSON key file.	`gcloud auth activate-service-account --key-file=$HOME/keys/key.json`
Auth	`gcloud auth login`	2		Obtains new user credentials through a browser flow so that gcloud can act as your Google account.	`gcloud auth login`
Compute	`gcloud compute instances list`	2	yes	Lists the VM instances in the project.	`gcloud compute instances list --project=agremotesensing`
Compute	`gcloud compute ssh`	2	yes	Opens an SSH session to a VM instance and manages the SSH keys for you.	`gcloud compute ssh ag-worker-1 --zone=us-central1-a`
Config	`gcloud config configurations activate`	2	yes	Switches to a different named configuration so that its project and account settings take effect.	`gcloud config configurations activate agremote`
Config	`gcloud config configurations list`	2	yes	Lists all named configurations and shows which one is active.	`gcloud config configurations list`
Config	`gcloud config get-value`	2		Prints the value of a single configuration property, which	`PROJECT_ID=$(gcloud config`

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				is useful in shell scripts.	`get-value core/project)`
Config	`gcloud config set account`	2		Switches the active credentialed account used by gcloud commands.	`gcloud config set account 'marron.bruce.mx@gmail.com'`
Config	`gcloud config set compute/region`	2		Sets the default region used when creating regional resources.	`gcloud config set compute/region us-central1`
Config	`gcloud config set compute/zone`	2	yes	Sets the default zone used when creating zonal resources such as VM instances.	`gcloud config set compute/zone us-central1-a`
Config	`gcloud init`	2		Interactively logs you in and sets a default project, account and region, creating or reinitializing a configuration.	`gcloud init --skip-diagnostics`
Help	`gcloud [COMMAND] --help`	2		Displays the full manual page for a command group or command, including its flags and examples.	`gcloud storage cp --help`
IAM	`gcloud iam service-accounts list`	2		Lists all service accounts in the active or specified project.	`gcloud iam service-accounts list --project=agremotesensing`
IAM	`gcloud projects add-iam-policy-binding`	2		Grants a role on the project to a user, group or service account.	`gcloud projects add-iam-policy-binding agremotesensing --member="serviceAccount:bmarron18-ag@agremotesensing.iam.gserviceaccount.com" --role="roles/earthengine.writer"`
IAM	`gcloud projects get-iam-policy`	2		Returns the project IAM policy (YAML by default), which can be flattened and filtered to show the roles held by one member.	`gcloud projects get-iam-policy agremotesensing --flatten="bindings[].members" --filter="bindings.members:marron.bruce.mx@gmail.com" --format="table(bindings.role)"`
IAM	`gcloud storage buckets add-iam-policy-binding`	2		Grants a role on a single bucket, such as the object write access that Earth Engine exports need.	`gcloud storage buckets add-iam-policy-binding`

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					`gs://ag-remote-bucket01 --member="user:marron.bruce.mx@gmail.com" --role="roles/storage.objectAdmin"`
IAM	`gcloud storage buckets get-iam-policy`	2		Displays the IAM policy bindings on a bucket so you can verify who has access to it.	`gcloud storage buckets get-iam-policy gs://ag-remote-bucket01`
Projects	`gcloud projects list`	2		Lists all projects that your account has permission to view.	`gcloud projects list`
Services	`gcloud services enable`	2		Enables one or more APIs, such as Earth Engine, Cloud Storage or Compute, on a project.	`gcloud services enable earthengine.googleapis.com storage.googleapis.com --project=agremotesensing`
Services	`gcloud services list --enabled`	2	yes	Lists the APIs currently enabled on the project.	`gcloud services list --enabled --project=agremotesensing`
Storage	`gcloud storage buckets create`	2		Creates a new bucket with a globally unique name in a chosen location.	`gcloud storage buckets create gs://ag-remote-bucket01 --location=us-central1 --uniform-bucket-level-access`
Storage	`gcloud storage buckets list`	2		Lists all buckets in the active project.	`gcloud storage buckets list --project=agremotesensing`
Storage	`gcloud storage cp -r`	2		Recursively copies an entire folder between a bucket and the local machine in either direction.	`gcloud storage cp -r gs://ag-remote-bucket01/exports ~/data/`
Storage	`gcloud storage du`	2	yes	Reports the storage used by objects or prefixes, which helps you stay within the 5 GiB free tier.	`gcloud storage du -s -r gs://ag-remote-bucket01`
Storage	`gcloud storage ls -l`	2	yes	Lists objects with their size and creation time, with an optional total.	`gcloud storage ls -l gs://ag-remote-bucket01/`
Storage	`gcloud storage mv`	2	yes	Moves or renames objects within or between buckets.	`gcloud storage mv gs://ag-remote-bucket01/exports/test_rgb.tif gs://ag-remote-bucket01/archive/`
Storage	`gcloud storage rm`	2		Deletes a single object or a wildcard-matched set of	`gcloud storage rm gs://ag-`

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				objects from a bucket.	`remote-bucket01/t.txt`
Storage	`gcloud storage rsync`	2	yes	Synchronizes the contents of a local directory and a bucket prefix.	`gcloud storage rsync -r gs://ag-remote-bucket01/exports ~/data/exports`
Auth	`gcloud auth application-default login`	3		Creates local Application Default Credentials that client libraries such as Python can use (the source notes this is avoided in this project).	`gcloud auth application-default login`
Auth	`gcloud auth application-default set-quota-project`	3	yes	Sets the project that Application Default Credentials bill and attribute quota to.	`gcloud auth application-default set-quota-project agremotesensing`
Auth	`gcloud auth print-access-token`	3	yes	Prints an OAuth2 access token for the active account, which you can use in REST API calls.	`curl -H "Authorization: Bearer $(gcloud auth print-access-token)" https://earthengine.googleapis.com/v1/projects/agremotesensing/assets`
Auth	`gcloud auth revoke`	3	yes	Revokes the credentials for an account and removes them from the local machine.	`gcloud auth revoke marron.bruce.mx@gmail.com`
Billing	`gcloud billing projects describe`	3	yes	Shows whether billing is enabled for a project and which billing account it is linked to.	`gcloud billing projects describe agremotesensing`
Components	`gcloud components install`	3		Installs an optional gcloud component.	`gcloud components install beta`
Components	`gcloud components list`	3	yes	Lists all components and shows whether each one is installed and up to date.	`gcloud components list`
Components	`gcloud components remove`	3		Uninstalls an installed gcloud component.	`gcloud components remove beta`
Components	`gcloud components update`	3		Updates the gcloud CLI and its installed components to the latest version.	`gcloud components update`
Components	`gcloud components update --version`	3		Reverts or pins the gcloud CLI to a specific earlier version.	`gcloud components update --version 591.0.0`
Compute	`gcloud compute instances create`	3	yes	Creates a new VM instance.	`gcloud compute instances create ag-worker-1 --zone=us-central1-a --machine-type=e2-medium`
Compute	`gcloud compute project-info describe`	3		Displays Compute Engine project metadata, quotas and	`gcloud compute project-info describe --`

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				the default region and zone.	`project=agremotesensing`
Compute	`gcloud compute regions list`	3	yes	Lists the available Compute Engine regions.	`gcloud compute regions list`
Config	`gcloud config configurations create`	3	yes	Creates a new named configuration to hold a separate set of project and account settings.	`gcloud config configurations create agremote`
Config	`gcloud config unset`	3	yes	Removes a property from the active configuration.	`gcloud config unset compute/zone`
Config	`gcloud info`	3		Prints diagnostic details about the local installation, including paths, the active account, the active project and the Python version.	`gcloud info`
Help	`gcloud help`	3	yes	Searches the gcloud help text for commands that match a term.	`gcloud help -- iam policy`
IAM	`gcloud iam roles describe`	3	yes	Shows the permissions included in a predefined or custom IAM role.	`gcloud iam roles describe roles/earthengine.writer`
IAM	`gcloud iam service-accounts create`	3	yes	Creates a new service account in the project.	`gcloud iam service-accounts create bmarron18-ag --display-name="Ag remote sensing SA" --project=agremotesensing`
IAM	`gcloud iam service-accounts keys create`	3		Creates a new private key for a service account and saves it locally as a JSON file.	`gcloud iam service-accounts keys create ~/keys/key.json --iam-account=bmarron18-ag@agremotesensing.iam.gserviceaccount.com`
IAM	`gcloud iam service-accounts keys list`	3	yes	Lists the keys that exist for a service account, which helps with auditing and rotation.	`gcloud iam service-accounts keys list --iam-account=bmarron18-ag@agremotesensing.iam.gserviceaccount.com`
IAM	`gcloud projects remove-iam-policy-binding`	3	yes	Revokes a role from a member at the project level.	`gcloud projects remove-iam-policy-binding agremotesensing --member="user:marron.bruce.mx@gmail.com" --role="roles/storage.admin"`

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IAM	`gcloud storage buckets remove-iam-policy-binding`	3	yes	Revokes a role from a member on a specific bucket.	`gcloud storage buckets remove-iam-policy-binding gs://ag-remote-bucket01 --member="user:marron.bruce.mx@gmail.com" --role="roles/storage.objectAdmin"`
Logging	`gcloud logging logs list`	3	yes	Lists the log names that exist in the project.	`gcloud logging logs list --project=agremotesensing`
Logging	`gcloud logging read`	3		Reads log entries that match a filter and prints them to the terminal.	`gcloud logging read "resource.type=gce_instance" --limit=10`
Projects	`gcloud projects create`	3	yes	Creates a new Google Cloud project.	`gcloud projects create agremotesensing --name="ag-remote-01"`
Projects	`gcloud projects delete`	3		Schedules a project for deletion, after which it can still be restored for 30 days.	`gcloud projects delete agremotesensing`
Projects	`gcloud projects describe`	3		Shows project metadata, including its name, project ID, project number and lifecycle state.	`gcloud projects describe agremotesensing`
Projects	`gcloud projects undelete`	3	yes	Restores a project that is pending deletion.	`gcloud projects undelete agremotesensing`
Services	`gcloud services disable`	3	yes	Disables an API on the project.	`gcloud services disable translate.googleapis.com --project=agremotesensing`
Services	`gcloud services list --available`	3		Lists every API that can be enabled on Google Cloud.	`gcloud services list --available --filter="name:earthengine"`
Storage	`gcloud storage buckets delete`	3	yes	Deletes an empty bucket.	`gcloud storage buckets delete gs://ag-remote-bucket01`
Storage	`gcloud storage buckets describe`	3	yes	Shows bucket metadata such as location, storage class and access settings.	`gcloud storage buckets describe gs://ag-remote-bucket01`
Storage	`gcloud storage cat`	3	yes	Prints the contents of an object to standard output.	`gcloud storage cat gs://ag-remote-bucket01/exports/fields.csv`
Storage	`gcloud storage objects describe`	3	yes	Shows the metadata for a single object, including its size,	`gcloud storage objects describe gs://ag-remote-`

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				content type and hashes.	`bucket01/exports/test_rgb.tif`
Storage	`gcloud storage rm -r`	3		Permanently deletes a folder prefix and all of its contents from a bucket.	`gcloud storage rm -r gs://ag-remote-bucket01/exports/old`